Arista Access Node MALL-ACC-80 QoS Congestion Event Resolution SOP

Node: MALL-ACC-80 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_6744

Description

The QoS congestion event failure on Arista Access Node MALL-ACC-80 is primarily caused by microbursts and interface tail drops resulting from inadequate Quality of Service (QoS) configuration. This can be attributed to the insufficiency of the configured QoS queues to handle the bursty traffic, ultimately leading to QoS queue starvation. Furthermore, the lack of proper policing and shaping mechanisms may exacerbate the congestion event, causing packet drops and service degradation. This situation is exacerbated by the fact that the Arista Access Node MALL-ACC-80 operates in a multi-tenancy environment where competing traffic patterns may further compound the congestion. As a result, it is essential to troubleshoot and rectify the QoS configuration to ensure optimal network performance and prevent service disruptions.

Pre-requisite Checks

- Pre-requisite: Verify if redundant link is available and functioning properly

- Pre-requisite: Ensure remote hands are dispatched to Malleswaram Node for physical intervention if required

- Pre-requisite: Verify if QoS configuration is up-to-date and aligns with the latest network policies

Potential Root Causes

Cause	Probability
Inadequate QoS queue configuration	High
Lack of policing and shaping mechanisms	Medium
Insufficient buffer size	Low

Diagnostic Commands & Expected Outputs

```
Command: show traffic statistics interface Ethernet1/1, Expected Output: Traffic statistics for interface Ethernet1/1: packets sent: 100000, packets received: 90000, packet drops: 10000, Description: This command is used to troubleshoot interface traffic statistics and identify potential congestion points.

Command: show qos queue summary, Expected Output: QoS queue summary: queues configured: 5, queues active: 3, queue drops: 1000, Description: This command is used to verify the current QoS queue configuration and identify potential queue starvation.

Command: show traffic policer summary, Expected Output: Traffic policer summary: policers configured: 2, policers active: 1, policer
```

```
drops: 500, Description: This command is used to verify the current policing configuration and identify potential policing issues.
```

Resolution Steps

Step 1: Step: Verify and update QoS queue configuration, Command: configure qos queue Ethernet1/1 queue 1 bandwidth 100mb, Description: Update QoS queue configuration to ensure adequate bandwidth allocation for the interface.

Step 2: Step: Configure policing and shaping mechanisms, Command: configure traffic policer Ethernet1/1 policer 1 rate 10mb, Description: Configure policing and shaping mechanisms to prevent excessive traffic from overwhelming the interface.

Step 3: Step: Verify buffer size, Command: show buffer size, Description: Verify buffer size to ensure it is adequate for the expected traffic pattern.

Post-Check Verification

```
Command: show traffic statistics interface Ethernet1/1, Expected Output: Traffic statistics for interface Ethernet1/1: packets sent: 100000, packets received: 90000, packet drops: 0, Description: Verify that packet drops have been eliminated after resolution.

Command: show qos queue summary, Expected Output: QoS queue summary: queues configured: 5, queues active: 3, queue drops: 0, Description: Verify that queue starvation has been eliminated after resolution.

Command: show traffic policer summary, Expected Output: Traffic policer summary: policers configured: 2, policers active: 1, policer drops: 0, Description: Verify that policing issues have been eliminated after resolution.
```

Roll-back Procedures

- Step: Revert QoS queue configuration, Command: configure qos queue Ethernet1/1 queue 1 bandwidth 50mb, Description: Revert QoS queue configuration to its previous state in case the new configuration causes issues.

- Step: Disable policing and shaping mechanisms, Command: configure traffic policer Ethernet1/1 policer 1 rate 5mb, Description: Disable policing and shaping mechanisms to revert to the previous configuration in case of issues.